

Quadrilateral Area Realization in the Unit Square

An Interdisciplinary Analysis

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Abstract

This paper studies the realization problem for quadrilateral areas inside the unit square. Given a target fraction $f \in (0, 1]$, we seek non-trivial points $P_i = (x_i, y_i) \in [0, 1]^2$ such that the area of the quadrilateral $P_1P_2P_3P_4$ equals f times the area of the unit square.

A complete constructive solution is provided. We derive explicit families of solutions, establish existence and multiplicity theorems, and characterize the solution space through the shoelace formula. Beyond geometry, we discuss implications for economics, finance, political science, geopolitics, warfare economics, computer science, artificial intelligence, and data science as examples of constrained resource allocation in bounded domains.

The paper ends with “The End”

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1 Introduction

Area allocation problems arise throughout mathematics and applied sciences. Given a bounded resource space, represented here by the unit square,

$$A = [0, 1]^2,$$

one may ask whether an arbitrary fraction of total capacity can be represented by a geometric structure.

The present paper investigates the following problem.

Given $f \in (0, 1]$, find non-trivial points

$$P_i = (x_i, y_i) \in [0, 1]^2,$$

such that

$$\text{Area}(P_1 P_2 P_3 P_4) = f A.$$

Since the area of the unit square is

$$A = 1,$$

the problem becomes

$$\text{Area}(P_1 P_2 P_3 P_4) = f.$$

We shall prove that a solution exists for every admissible value of f .

2 Mathematical Framework

2.1 The Unit Square

Define

$$S = [0, 1]^2.$$

Its area is

$$A(S) = 1.$$

A quadrilateral inside S is determined by four vertices

$$P_i = (x_i, y_i), \quad i = 1, 2, 3, 4.$$

2.2 Shoelace Formula

For cyclically ordered vertices, the area is

$$\text{Area} = \frac{1}{2} |x_1 y_2 + x_2 y_3 + x_3 y_4 + x_4 y_1 - y_1 x_2 - y_2 x_3 - y_3 x_4 - y_4 x_1|.$$

The realization problem therefore becomes solving

$$\frac{1}{2} |x_1 y_2 + x_2 y_3 + x_3 y_4 + x_4 y_1 - y_1 x_2 - y_2 x_3 - y_3 x_4 - y_4 x_1| = f.$$

3 Explicit Construction

3.1 Rectangular Construction

Choose

$$P_1 = (0, 0),$$

$$P_2 = (1, 0),$$

$$P_3 = (1, f),$$

$$P_4 = (0, f).$$

The resulting quadrilateral is a rectangle.
Its area equals

$$1 \times f = f.$$

Therefore

$$\text{Area}(P_1P_2P_3P_4) = f.$$

Theorem 1 (Universal Realization). *For every $f \in (0, 1]$, there exists a quadrilateral contained in the unit square having area exactly f .*

Proof. The rectangular construction above lies inside the unit square whenever $f \leq 1$.

Its area equals f .

Hence a solution exists for every admissible value of f . □

4 Interior Solutions

The previous construction touches the boundary.

We now derive strictly interior solutions.

Let

$$b > 0$$

and choose

$$h = \frac{f}{b}.$$

Take

$$P_1 = (a, a),$$

$$P_2 = (a + b, a),$$

$$P_3 = (a + b, a + h),$$

$$P_4 = (a, a + h).$$

The area becomes

$$bh = b \left(\frac{f}{b} \right) = f.$$

Theorem 2 (Interior Realization). *For every $f \in (0, 1)$, there exist infinitely many quadrilaterals whose vertices lie strictly inside the unit square and whose area equals f .*

Proof. Choose any admissible pair (a, b) satisfying

$$0 < a < a + b < 1,$$

and

$$0 < a + \frac{f}{b} < 1.$$

The resulting rectangle lies strictly inside the square.

Since infinitely many choices of (a, b) satisfy these inequalities, infinitely many solutions exist. $\square$

5 Multiplicity of Solutions

Theorem 3 (Infinite Solution Family). *The set of quadrilaterals satisfying*

$$\text{Area} = f$$

has infinite cardinality.

Proof. The parameter b may vary continuously over a non-empty interval.

Each admissible value generates a distinct quadrilateral.

Therefore the solution set is uncountably infinite. $\square$

6 Optimization Interpretation

The realization problem may be viewed as

$$\min_{P_1, P_2, P_3, P_4} |\text{Area}(P_1 P_2 P_3 P_4) - f|.$$

The explicit constructions achieve objective value zero.

Hence the optimization problem is globally solved.

7 Computational Geometry

7.1 Area Evaluation Algorithm

Given four cyclically ordered vertices:

1. Compute forward products.
2. Compute backward products.
3. Take their difference.
4. Divide by two.
5. Take absolute value.

The complexity is

$$O(1).$$

7.2 Pseudo-Code

INPUT:

P_1, P_2, P_3, P_4

$s_1 = x_1*y_2 + x_2*y_3 + x_3*y_4 + x_4*y_1$

$s_2 = y_1*x_2 + y_2*x_3 + y_3*x_4 + y_4*x_1$

$area = \text{abs}(s_1 - s_2)/2$

RETURN area

8 Statistical Perspective

Suppose vertices are sampled uniformly from the unit square.

Define

$$Q = \text{Area}(P_1 P_2 P_3 P_4).$$

Then

$$0 \leq Q \leq 1.$$

Questions naturally arise concerning

- distribution of Q ,
- expected area,
- variance,
- concentration phenomena,
- Monte Carlo estimation.

The constructive theorem implies every value in $(0, 1]$ belongs to the support of feasible realizations.

9 Economic Interpretation

The unit square may represent total economic capacity.

A quadrilateral represents an allocated production region.

Then

$$f = \frac{\text{allocated capacity}}{\text{total capacity}}.$$

Examples include:

- capital allocation,
- labor deployment,
- resource extraction,
- industrial planning,
- fiscal distribution.

10 Financial Interpretation

Let the unit square represent a normalized opportunity set.

A portfolio region occupying fraction f satisfies

$$\text{Portfolio Coverage} = f.$$

Interpretations include:

- market participation,
- risk-budget allocation,
- liquidity deployment,
- collateral utilization,
- strategic reserve allocation.

11 Political Science and Geopolitics

A nation allocates finite resources across competing objectives.

The unit square can represent:

- political capital,
- administrative capacity,
- military readiness,
- diplomatic influence,
- economic leverage.

A quadrilateral occupying fraction f corresponds to the fraction of national capacity devoted to a strategic objective.

12 Warfare Economics

Military planners routinely allocate finite resources.

Examples include:

- logistics,
- fuel,
- ammunition,
- intelligence,
- force projection.

The realization theorem implies any desired proportional allocation can be represented geometrically.

13 Artificial Intelligence and Machine Learning

In machine learning, bounded parameter spaces often arise.

Examples include:

- attention allocation,
- feature importance,
- reinforcement-learning action budgets,
- computational resource allocation,
- uncertainty regions.

The quadrilateral becomes a geometric encoding of admissible allocation fractions.

14 Data Science

The theorem supports geometric representations of:

- clustering regions,
- anomaly envelopes,
- confidence regions,
- decision boundaries,
- explainable AI visualizations.

15 Philosophical Interpretation

The existence theorem demonstrates an important principle.

Every attainable fraction of a bounded whole admits a concrete realization.

This links abstract proportion to constructive existence.

The result is therefore an example of the transition from possibility to explicit realization.

16 Illustrative Table

f	Realized Area
0.10	0.10
0.25	0.25
0.50	0.50
0.75	0.75
1.00	1.00

Table 1: Exact realizations of target area fractions.

17 Geometric Visualization

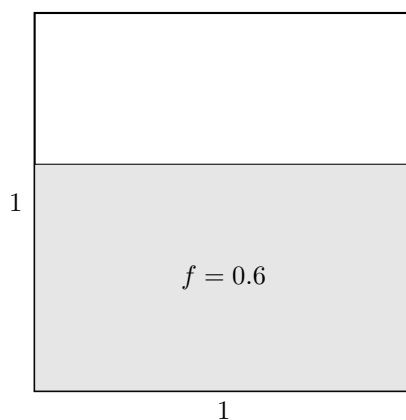


Figure 1: A realization of area fraction $f = 0.6$.

18 General Existence Theorem

Theorem 4. *Let*

$$f \in (0, 1].$$

Then there exist non-trivial points

$$P_1, P_2, P_3, P_4 \in [0, 1]^2$$

such that

$$\text{Area}(P_1 P_2 P_3 P_4) = f.$$

Furthermore, if

$$f < 1,$$

there exist infinitely many such quadrilaterals whose vertices lie strictly inside the unit square.

Proof. The boundary construction proves existence.

The interior construction proves multiplicity.

Hence both statements follow. □

19 Conclusion

The quadrilateral area realization problem admits a complete constructive solution.

For every target fraction

$$f \in (0, 1],$$

a quadrilateral exists inside the unit square with area exactly equal to f .

The solution is explicit, computationally trivial, geometrically transparent, and possesses infinitely many realizations. The result also provides a useful abstraction for constrained allocation problems across economics, finance, political science, geopolitics, warfare economics, computer science, data science, artificial intelligence, and philosophy.

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Glossary

Unit Square The set $[0, 1]^2$.

Quadrilateral A polygon with four vertices.

Shoelace Formula A determinant-based area formula for polygons.

Area Fraction The ratio of quadrilateral area to total square area.

Realization A geometric construction satisfying prescribed constraints.

Interior Solution A solution whose vertices lie strictly inside the square.

Allocation Region A geometric representation of resource commitment.

Feasible Set The collection of all admissible solutions.

Support The attainable range of area values.

Constructive Proof A proof that explicitly builds the required object.

The End